YASKAWA

YRC1000micro OPTIONS
INSTRUCTIONS

REFERENCE MANUAL
FOR PROGRAMMING PENDANT CUSTOMIZATION FUNCTION
(MotoConnectCSU.dll API SPECIFICATIONS
FOR DEVELOPING Visual Studio. NET C#)

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

MOTOMAN INSTRUCTIONS

MOTOMAN-□□□ INSTRUCTIONS
YRC1000micro INSTRUCTIONS
YRC1000micro OPERATOR'S MANUAL
YRC1000micro MAINTENANCE MANUAL
YRC1000micro ALARM CODES (MAJOR ALARMS) (MINOR ALARMS)

The YRC1000micro alarm codes above consists of "MAJOR ALARMS" and "MINOR ALARMS".

Please have the following information available when contacting Yaskawa Customer Support:
- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
  Robot Serial Number *(Located on robot data plate)*
  Robot Sales Order Number *(Located on controller data plate)*

Part Number: 181298-1CD
Revision: 0

Yaskawa Customer Support
- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

MANUAL NO.
HW1484522

DANGER

- This manual explains the programming pendant customization function of the YRC1000micro system. Read this manual carefully and be sure to understand its contents before handling the YRC1000micro. Any matter not described in this manual must be regarded as “prohibited” or “improper”.
- General information related to safety are described in “Chapter 1. Safety” of the YRC1000micro INSTRUCTIONS. To ensure correct and safe operation, carefully read “Chapter 1. Safety” of the YRC1000micro INSTRUCTIONS.

CAUTION

- In some drawings in this manual, protective covers or shields are removed to show details. Make sure that all the covers or shields are installed in place before operating this product.
- YASKAWA is not responsible for incidents arising from unauthorized modification of its products. Unauthorized modification voids the product warranty.

NOTICE

- The drawings and photos in this manual are representative examples and differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice when necessary due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- If your copy of the manual is damaged or lost, contact a YASKAWA representative to order a new copy. The representatives are listed on the back cover. Be sure to tell the representative the manual number listed on the front cover.

Notes for Safe Operation

Read this manual carefully before installation, operation, maintenance, or inspection of the YRC1000micro.

In this manual, the Notes for Safe Operation are classified as “DANGER”, “WARNING”, “CAUTION”, or “NOTICE”.

DANGER

Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. Safety Signs identified by the signal word DANGER should be used sparingly and only for those situations presenting the most serious hazards.

WARNING

Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury. Hazards identified by the signal word WARNING present a lesser degree of risk of injury or death than those identified by the signal word DANGER.

CAUTION

Indicates a hazardous situation, which if not avoided, could result in minor or moderate injury. It may also be used without the safety alert symbol as an alternative to “NOTICE”.

NOTICE

NOTICE is the preferred signal word to address practices not related to personal injury. The safety alert symbol should not be used with this signal word. As an alternative to “NOTICE”, the word “CAUTION” without the safety alert symbol may be used to indicate a message not related to personal injury.

Even items described as “CAUTION” may result in a serious accident in some situations.
At any rate, be sure to follow these important items.

NOTE

To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” and “CAUTION".

DANGER

- Before operating the manipulator, make sure the servo power is turned OFF by performing the following operations. When the servo power is turned OFF, the SERVO ON LED on the programming pendant is turned OFF.
  - Press the emergency stop button on the programming pendant or on the external control device, etc.
  - Disconnect the safety plug of the safety fence. (when in the play mode or in the remote mode)

If operation of the manipulator cannot be stopped in an emergency, personal injury and/or equipment damage may result.

Fig. : Emergency Stop Button

- Before releasing the emergency stop, make sure to remove the obstacle or error caused the emergency stop, if any, and then turn the servo power ON.

Failure to observe this instruction may cause unintended movement of the manipulator, which may result in personal injury.

Fig. : Release of Emergency Stop

- Observe the following precautions when performing a teaching operation within the manipulator's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence. In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the manipulator’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this instruction may cause improper or unintended movement of the manipulator, which may result in personal injury.

- Confirm that no person is present in the manipulator's operating range and that the operator is in a safe location before:
  - Turning ON the YRC1000micro power
  - Moving the manipulator by using the programming pendant
  - Running the system in the check mode
  - Performing automatic operations

Personal injury may result if a person enters the manipulator's operating range during operation. Immediately press an emergency stop button whenever there is a problem. The emergency stop button is located on the right of the programming pendant.

- Read and understand the Explanation of the Warning Labels before operating the manipulator.

DANGER

- In the case of not using the programming pendant, be sure to supply the emergency stop button on the equipment. Then before operating the manipulator, check to be sure that the servo power is turned OFF by pressing the emergency stop button.
  Connect the external emergency stop button to the 4-14 pin and 5-15 pin of the Safety connector (Safety).
- Upon shipment of the YRC1000micro, this signal is connected by a jumper cable in the dummy connector. To use the signal, make sure to supply a new connector, and then input it.

If the signal is input with the jumper cable connected, it does not function, which may result in personal injury or equipment damage.

WARNING

- Perform the following inspection procedures prior to conducting manipulator teaching. If there is any problem, immediately take necessary steps to solve it, such as maintenance and repair.
  - Check for a problem in manipulator movement.
  - Check for damage to insulation and sheathing of external wires.
- Return the programming pendant to a safe place after use.

If the programming pendant is left unattended on the manipulator, on a fixture, or on the floor, etc., the Enable Switch may be activated due to surface irregularities of where it is left, and the servo power may be turned ON. In addition, in case the operation of the manipulator starts, the manipulator or the tool may hit the programming pendant left unattended, which may result in personal injury and/or equipment damage.

Definition of Terms Used Often in This Manual

The MOTOMAN is the YASKAWA industrial robot product.

The MOTOMAN usually consists of the manipulator, the YRC1000micro controller, manipulator cables, the YRC1000micro programming pendant (optional), and the YRC1000micro programming pendant dummy connector (optional).

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
YRC1000micro controller	YRC1000micro
YRC1000micro programming pendant	Programming pendant (optional)
Cable between the manipulator and the controller	Manipulator cable
YRC1000micro programming pendant dummy connector	Programming pendant dummy connector (optional)

Descriptions of the programming pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys /Symbol Keys	The keys which have characters or symbols printed on them are denoted with [ ]. ex. [ENTER]
	Axis Keys /Number Keys	[Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, ex. [SHIFT]+[COORD]
	Mode Key	Three kinds of modes that can be selected by the mode key are denoted as follows: REMOTE, PLAY, or TEACH
	Button	Three buttons on the upper side of the programming pendant are denoted as follows: HOLD button START button EMERGENCY STOP button
	Displays	The menu displayed in the programming pendant is denoted with { }. e.g. {JOB}
PC Keyboard		The name of the key is denoted. e.g. Ctrl key on the keyboard

Description of the Operation Procedure

In the explanation of the operation procedure, the expression "Select • • • " means that the cursor is moved to the object item and [SELECT] is pressed, or that the item is directly selected by touching the screen.

Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or brand names for each company or corporation. The indications of (R) and TM are omitted.

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1 Outline

In this manual, details of APIs used when the programming pendant customization application is developed with Visual Studio 2008 C# are explained.

2 LIBRARY OPEN-CLOSE API

MotoConnect

Constructor. Execute initialization of the library.

- **Syntax: Default IP address**
  (When Controller 10.0.0.2/ Pendant 10.0.0.4)
  MotoConnect (bool Endian)

- **Parameter**
  [in]*Endian*

  Endian type of the pendant CPU

Value	Description
true	Big endian
false	Little endian
Set “false” when the controller is the YRC1000micro.	

- **Return Value**
  No value

- **Syntax: Other than default IP address**
  MotoConnect(bool Endian, string Ycplp, string Ypplp)

- **Parameter**
  [in]*Endian*

  Endian type of the pendant CPU

Value	Description
true	Big endian
false	Little endian
Set “false” when the controller is the YRC1000micro.	

[in]*Ycplp*

Controller IP address

[in]*Ypplp*

pendant IP address

- **Return Value**
  No value

3 SYSTEM MONITOR API

YppGetVarData

Acquires the value of variable (B, I, D, R).

- **Syntax**
  int YppGetVarData(ushort DataType, ushort DataIndex, out int rData)

- **Parameter**
  [in] *DataType*

  Variable type

Value	Description
1	B variable
2	I variable
3	D variable
4	R variable

[in] *DataIndex*

Variable number

[out] *rData*

Variable value (result of acquisition)

- **Return value**
  0 : Normal end

  Other than 0 : Error

YppReadIO

Reads I/O.

■ **Syntax**
int YppReadIO(uint Address, out ushort Value)

■ **Parameter**
[in] *Address*

Address of I/O to be read.

Value	Description
10 - 5127	Universal input #00010 - #05127 (4096)
10010 - 15127	Universal output #10010 - #15127 (4096)
20010 - 21287	External input #20010 - #21287 (1024)
30010 - 31287	External output #30010 - #31287 (1024)
40010 - 42567	Specific input #40010 - #42567 (2048)
50010 - 55127	Specific output #50010 - #55127 (4096)
60010 - 60647	I/F panel #60010 - #60647 (512)
70010 - 79997	Auxiliary relay #70010 - #79997 (7992)
80010 - 85127	Control input #80010 - #85127 (4096)
87010 - 87207	Pseud input #87010 - #82207 (160)
27010 - 29567	Network input #27010 - #29567 (2048)
37010 - 39567	Network output #37010 - #39567 (2048)
1000000 - 1000999	Universal register M000 - M559 (560) Analog output register M560 - M599 (40) Analog input register M600 - M639 (40) System register M640 - M999 (360)

[out]Value

I/O value (result of acquisition)

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetPosVarData

Acquires the position-type variable.

■ **Syntax**
int YppGetPosVarData
(ushort DataType, ushort DataIndex, out int[] rData)

■ **Parameter**
[in]*DataType*

Variable type

Value	Description
5	Robot
6	Base
7	Station

[in]*DataIndex*

Variable number

[out]*rData (int[10])*

Position data (result of acquisition) Note:Area for int[10] is required.

Array	Description	
	Bit No.	Content
rData[0]	D05 - D00	Variable type 0 Pulse 16 Cartesian (base coordinates) 17 Cartesian (robot coordinates) 18 Cartesian (tool coordinates) 19 Cartesian (user coordinates) 20 Cartesian (reserved for master tool)
	D07 - D06	Reserved by manufacturer
	D08	0:Front 1:Back
	D09	0:Upper arm 1: Lower arm
	D10	0:Flip 1 :No flip
	D11	0:R<180deg 1:R>=180deg
	D12	0:T<180deg 1:T>=180deg
	D13	0:S<180de 1:S>=180deg
	D14 - D15	Reserved by manufacturer
	D16 - D21	Tool number (0 - 23)
	D22 - D27	User coordinate number
	D28 - D31	Reserved by manufacturer
rData[1]		(Extended attribute)

Array	Description	
	Bit No.	**Content**
Array	**Pulse**	**Cartesian**
rData[2]	1st axis (S) pulse value	X-axis coordinate (unit: micron)
rData[3]	2nd axis (L) pulse value	Y-axis coordinate (unit: micron)
rData[4]	3rd axis (U) pulse value	Z-axis coordinate (unit: micron)
rData[5]	4th axis (R) pulse value	Wrist angle Rx (unit: 0.0001deg)
rData[6]	5th axis (B) pulse value	Wrist angle Ry (unit: 0.0001deg)
rData[7]	6th axis (T) pulse value	Wrist angle Rz (unit: 0.0001deg)
rData[8]	7th axis (E) pulse value	angle Re (unit: 0.0001deg)
rData[9]	8th axis pulse value	8th axis pulse value (micron in the case of traveling axis)

■ **Return value**

0 : Normal end

Other than 0 : Error

YppGetAlarmStatus

Acquires the error and alarm status.

■ **Syntax**
int YppGetAlarmStatus(out short Status)

■ **Parameter**
[out]*Status*

Error and alarm status. (result of acquisition)

Value	Description
D00	1: In error status
D01	1: In alarm status
D02 - D15	(Unused)

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetAlarmCode

Acquires the error and alarm code.

■ **Syntax**
int YppGetAlarmCode(out ushort ErrorNo, out ushort ErrorData, out ushort AlarmNum, out ushort[] AlarmNo, out ushort[] AlarmData)

■ **Syntax**
[out] *ErrorNo*

Error number (result of acquisition)

[out*] ErrorData*

Error data (result of acquisition)

[out] *AlarmNum*

The numbers of alarm (result of acquisition) Note: 4 at maximum

[out] *AlarmNo (ushort[4])*

Alarm number (result of acquisition)

Note: The area for ushort[4] is required.

[out] *AlarmData (ushort[4])*

Alarm data (ushort[4]) (result of acquisition)

Note: The area for ushort[4] is required.

■ **Return value**

0	: Normal end
Other than 0	: Error

YppGetMode

Acquires the operation mode of the system.

■ **Syntax**
int YppGetMode(out short Mode, out short Remote)

■ **Parameter**
[out] *Mode*

Operation mode (result of acquisition)

Value	Description
1	Teach mode
2	Play mode

[out] *RemoteMode*

Command remote mode (result of acquisition)

Value	Description
0	Command remote OFF
1	Command remote ON

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetCycle

Acquires the cycle mode of the system.

■ **Syntax**
int YppGetCycle(out short Cycle)

■ **Parameter**
[out] *Cycle*

Cycle (result of acquisition)

Value	Description
0	Step
1	One Cycle
3	Auto

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetServoPower

Acquires the ON/OFF status of the servo power.

■ **Syntax**
int YppGetServoPower(out short Power)

■ **Parameter**
[out] *Power*

Cycle (result of acquisition)

Value	Description
0	Servo Power OFF
1	Servo Power ON

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetPlayStatus

Acquires the operation status of the job.

■ **Syntax**

int YppGetPlayStatus(out short Start, out short Hold)

■ **Parameter**

[out] *Start*

Operation status (result of acquisition)

Value	Description
0	Start OFF
1	Start ON

[out] *Hold*

Hold status (result of acquisition)

Value	Description
0	Hold OFF
1	Hold ON

■ **Return value**

0 : Normal end

Other than 0 : Error

YppGetMasterJob

Retrieves the master job name of the specified local task.

Syntax

int YppGetMasterJob(ushort TaskNo, out string JobName)

Parameter

[in] *TaskNo*

Task number

Value	Description
0	Master Task
1	SubTask 1
2	SubTask 2
3	SubTask 3
4	SubTask 4
5	SubTask 5

[out] *JobName*

Master job name (up to 32 characters for a job name)

Return value

0 : Normal end

Other than 0 : Error

YppGetCurrentJob

Acquires the name, line, step number of the current job.

■ **Syntax**

int YppGetCurrentJob(ushort TaskNo, out ushort LineNo, out ushort StepNo, out string JobName)

■ **Parameter**

[in] *TaskNo*

Task number

Value	Description
0	Master Task
1	SubTask 1
2	SubTask 2
3	SubTask 3
4	SubTask 4
5	SubTask 5

[out] *LineNo*

Job line number

[out] *StepNo*

Step number

[out] *JobName*

Job name (up to 32 characters available)

■ **Return value**

0 : Normal end

Other than 0 : Error

YppGetSpecialOpStatus

Retrieves the status of special operation.

■ **Syntax**
int YppGetSpecialOpStatus(out short SpecialOpStatus)

■ **Parameter**
[out] *SpecialOpStatus*

Special operation status (Result)

Value	Description
D0	Check operation
D1	Safety speed operation
D2	Dry-run speed operation
D3	Machine lock operation
D4	(Reserved)
D5	Low speed operation
D6	Weaving prohibited
D7-D15	(Reserved)

■ **Return value**
0 : Normal end

Other than 0 : Error

■ **Restrictions**
This API is valid only in Play mode.

YppGetJobDate

Acquires the job date.

■ **Syntax**

int YppGetJobData(string JobName, out short[] SysTimeData)

■ **Parameter**

[in] *JobName*

Job name (up to 32 characters available)

[out] *SysTimeData (ushort[6])*

Time and date data (result of acquisition)

Note: Area for short[6] is required.

SysTimeData[0]	:	Year
SysTimeData[1]	:	Month
SysTimeData[2]	:	Date
SysTimeData[3]	:	Time
SysTimeData[4]	:	Minute
SysTimeData[5]	:	Second

■ **Return value**

0	: Normal end
Other than 0	: Error

YppGetCartPosEx

Acquires the current position in Cartesian Coordinates with a specified coordinate frame (Robot, Base, User).

Syntax

int YppGetCartPosEx(short RobotNo, short Frame, short ToolNo, out int[] Pos, out short Config)

Parameter

[in] *RobotNo*

Robot number

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)

[in] *Frame*

Specified coordinate frame

Value	Description
0	Base coordinate
1	Robot coordinate
2, 3, ...	User coordinate1, 2, ...

[in] *ToolNo*

Tool number 0 to 63

[out] *Pos (int[12])*

Position data (result of acquisition)

Note: Area for int[12] is required.

Value	Description
IPos[0]	X-axis coordinate (unit: micron)
IPos[1]	Y-axis coordinate (unit: micron)
IPos[2]	Z-axis coordinate (unit: micron)
IPos[3]	Wrist angle Rx (unit: 0.0001deg)
IPos[4]	Wrist angle Ry (unit: 0.0001deg)
IPos[5]	Wrist angle Rz (unit: 0.0001deg)
IPos[6]	angle Re (unit: 0.0001deg)
IPos[7]	(Reserved)
IPos[8]	1st external axis pulse value (micron in the case of traveling axis)
IPos[9]	2nd external axis pulse value (micron in the case of traveling axis)
IPos[10]	3rd external axis pulse value (micron in the case of traveling axis)
IPos[11]	(Unused)

[out] *Config*

Figure information (result of acquisition)

Value	Description
D00	0:Front 1:Back
D01	0:Upper arm 1:Lower arm
D02	0:Flip 1:No flip
D03	0:R<180deg 1:R>=180deg
D04	0:T<180deg 1:T>=180deg
D05	0:S<180deg 1:S>=180deg
D06	0:L<0deg 1:L>=0deg
D07 -D15	Reserved

■ **Return value**

0 : Normal end

Other than 0 : Error

YppGetPulsePos

Acquires the current position in pulse.

Syntax

int YppGetPulsePos(uint CtrlGrp, out int[] Pos)

Parameter

[in] *CtrlGrp*

Control group

<YRC1000micro>

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[[out] *Pos (int[8])*

Pulse coordinate position data

Note: Area for int[8] is required;

Value	Description
lPos[0]	1st axis (S) pulse value
lPos[1]	2nd axis (L) pulse value
lPos[2]	3rd axis (U) pulse value
lPos[3]	4th axis (R) pulse value
lPos[4]	5th axis (B) pulse value
lPos[5]	6th axis (T) pulse value
lPos[6]	7th axis (E) pulse value
lPos[7]	8th axis pulse value

Return value

0 : Normal end

Other than 0 : Error

YppGetFBPulsePos

Retrieves the feedback position in pulse count.

Syntax

int YppGetFBPulsePos(uint CtrlGrp, out int[] Pos)

Parameter

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[[out] *Pos (int[8])*

Position coordinates in pulse (Result)

Note: Area for int[8] is required;

Value	Description
Pos[0]	1st axis (S) pulse value
Pos[1]	2nd axis (L) pulse value
Pos[2]	3rd axis (U) pulse value
Pos[3]	4th axis (R) pulse value
Pos[4]	5th axis (B) pulse value
Pos[5]	6th axis (T) pulse value
Pos[6]	7th axis (E) pulse value
Pos[7]	8th axis pulse value

Return value

0 : Normal end

Other than 0 : Error

YppGetServoSpeed

Retrieves the current servo speed by seconds.

Syntax

int YppGetServoSpeed(uint CtrlGrp, out int[] Speed)

Parameter

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[[out] *Speed (int[8])*

Servo speed data (Result)

Note: Area for int[8] is required;

Value	Description
Speed[0]	1st axis (S) direction (unit: pulses per second - pps)
Speed[1]	2nd axis (L) direction (unit: pps)
Speed[2]	3rd axis (U) direction (unit: pps)
Speed[3]	4th axis (R) direction (unit: pps)
Speed[4]	5th axis (B) direction (unit: pps)
Speed[5]	6th axis (T) direction (unit: pps)
Speed[6]	7th axis (E) direction (unit: pps)
Speed[7]	8th axis direction (unit: pps)

Return value

0 : Normal end

Other than 0 : Error

YppGetFBSpeed

Retrieves the feedback speed by seconds.

Syntax

int YppGetFBSpeed(uint CtrlGrp, out int[] Speed)

Parameter

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[[out] *Speed (int[8])*

Feedback speed data (Result)

Note: Area for int[8] is required;

Value	Description
Speed[0]	1st axis (S) direction (unit: pulses per second - pps)
Speed[1]	2nd axis (L) direction (unit: pps)
Speed[2]	3rd axis (U) direction (unit: pps)
Speed[3]	4th axis (R) direction (unit: pps)
Speed[4]	5th axis (B) direction (unit: pps)
Speed[5]	6th axis (T) direction (unit: pps)
Speed[6]	7th axis (E) direction (unit: pps)
Speed[7]	8th axis direction (unit: pps)

Return value

0 : Normal end

Other than 0 : Error

YppGetTorque

Retrieves the percentage to the maximum current servo torque value.

Syntax

int YppGetTorque(uint CtrlGrp, out int[] TorquePercent)

Parameter

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[[out] *TorquePercent (int[8])*

Torque percentage (Result)

Note: Area for int[8] is required;

Value	Description
TorquePercent[0]	1st axis (S) direction (unit: 0.01% of max torque)
TorquePercent[1]	2nd axis (L) direction (unit: 0.01%)
TorquePercent[2]	3rd axis (U) direction (unit: 0.01%)
TorquePercent[3]	4th axis (R) direction (unit: 0.01%)
TorquePercent[4]	5th axis (B) direction (unit: 0.01%)
TorquePercent[5]	6th axis (T) direction (unit: 0.01%)
TorquePercent[6]	7th axis (E) direction (unit: 0.01%)
TorquePercent[7]	8th axis direction (unit: 0.01%)

Return value

0 : Normal end

Other than 0 : Error

YppGetSystemTimes

Acquires the current system time.

Syntax

int YppGetSystemTime(short TimeType, out short[] SysTimeData, out int ElapsedTime)

Parameter

[in] *TimeType*

Type of the system time

Value	Description
0	Time during the power supply of the controller is ON
1	Time during the servo power supply is ON
2	Time during the play back operation
3	Moving time
4	Operation time

[out] *SysTimeData (short[6])*

Date and time data (result of acquisition)

Note: Area for short[6] is required.

SysTimeData[0] : Year
SysTimeData[1] : Month
SysTimeData[2] : Date
SysTimeData[3] : Time
SysTimeData[4] : Minute
SysTimeData[5] : Second

[out] *ElapsedTime*

Elapsed time (result of acquisition)

Return value

0 : Normal end

Other than 0 : Error

YppGetJogSpeed

Acquires the current jog speed.

■ **Syntax**

int YppGetJogSpeed(out short JogSpeed)

■ **Parameter**

[out] *JogSpeed*

Jog speed

Value	Description
0	Inching
1	Low speed
2	Middle speed
3	High speed
4	Maxinum speed

■ **Return value**

0 : Normal end

Other than 0 : Error

YppGetJogCoord

Acquires the current jog coordinates.

Syntax

int YppGetJogCoord(out short JogCoord)

Parameter

[out] *JogCoord*

Jog coordinates

Value	Description
0	Joint coordinates
1	Cartesian coordinates
2	Cylindrical coordinates
3	Tool coordinates
4	User coordinates
5	External reference point coordinates
6	Teaching line coordinates

Return value

0 : Normal end

Other than 0 : Error

YppGetSVarInfo

Acquires the value of S variable.

■ **Syntax**

int YppGetSVarInfo(ushort DataIndex, out string SVarData)

■ **Parameter**

[in] *DataIndex*

Parameter

[out] *SVarData*

S variable value (S variable: up to 32 characters) (result of acquisition)

■ **Return value**

0	: Normal end
Other than 0	: Error

YppGetJogCtrlGroup

Acquires the control group of the current jog operation target.

■ **Syntax:**
int YppGetJogCtrlGroup(out uint GroupKind, out uint GroupNo);

■ **Parameter**
[*out*] GroupKind

Control group kind

Value	Description
1	ROBOT
2	BASE
3	STATION

[*out*] GroupNo

Control group number

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetToolNo

Acquires the operation tool number to the specified control group.

■ **Syntax:**
int YppGetToolNo(uint RobotNo, out short ToolNo, out ushort ErrorNo);

■ **Parameter**
[*in*] RobotNo

Robot number for acquiring the tool number

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)

[*out*] ToolNo

Operation tool number

[*out*] ErrorNo

Error number

Value	Description
0x3370	Specified robot number error

■ **Return value**
0 : Normal end

Other than 0 : Error

4 SYSTEM CONTROL API

YppSetVarData

Sets the value of the variable (B, I, D, R).

■ **Syntax**
int YppSetVarData(ushort DataType, ushort DataIndex, int sData)

■ **Parameter**
[in] *DataType*

Variable type

Value	Description
1	B variable
2	I variable
3	D variable
4	R variable

[in] *DataIndex*

Variable number

[in] *sData*

Variable value (set value)

■ **Return value**
0 : Normal end

Other than 0 : Error

■ Relevant parameter

Number	Function	Setting value	Default
S2C541	Permission of variable and I/O writing during play mode	0: Permitted 1: Not permitted	1 (Not permitted)
S2C542	Permission of variable and I/O writing during edit lock	0: Permitted 1: Not permitted	1 (Not permitted)

NOTICE

When “0” is set to “S2C541”, writing operation during play back mode becomes valid. However, please operate the manipulator with caution since this operation may influence the cycle time.

NOTICE

The “edit lock status” to which S2C542 can specify are following status.

- In alarm status
- External memory is being used
- Data transmitting function is being used
- Specified input EDIT_LOCK (#40064) is turned ON.

YppWriteIO

Writes I/O.

■ **Syntax**
int YppWriteIO(uint Address, uint Value)

■ **Parameter**
[in] *Address*

Address of I/O

Value	Description
10010 - 15127	Universal output #10010 - #15127 (4096)
60010 - 60647	I/F panel #60010 - #60647 (512)
27010 - 29567	Network input #27010 - #29567 (2048)
1000000 - 1000559	Universal register M000 - M559 (560)

[in] *Value*

Inputting value

■ **Return value**
0 : Normal end

Other than 0 : Error

■ Relevant parameter

Number	Function	Setting value	Default
S2C541	Permission of variable and I/O writing during play mode	0: Permitted 1: Not permitted	1 (Not permitted)
S2C542	Permission of variable and I/O writing during edit lock	0: Permitted 1: Not permitted	1 (Not permitted)

NOTICE

When “0” is set to “S2C541”, writing operation during play back mode becomes valid. However, please operate the manipulator with caution since this operation may influence the cycle time.

NOTICE

The “edit lock status” to which S2C542 can specify are following status.

- In alarm status
- External memory is being used
- Data transmitting function is being used
- Specified input EDIT_LOCK (#40064) is turned ON.

YppSetPosVarData

Sets the position-type variable

■ **Syntax**
int YppSetPosVarData(ushort DataType, ushort DataIndex, int[] sData)

■ **Parameter**
[in] *DataType*

Variable type

Value	Description
5	Robot
6	Base
7	Station

[in] *DataIndex*

Variable number

[in] *sData (int[10])*

Position data (set value)

Note: ARea for int[10] is required.

Array	Description	
	Bit No.	**Content**
rData[0]	D05 - D00	Variable type 0 Pulse 16 Cartesian (base coordinates) 17 Cartesian (robot coordinates) 18 Cartesian (tool coordinates) 19 Cartesian (user coordinates) 20 Cartesian (reserved for master tool)
	D07 - D06	Reserved by manufacturer
	D08	0:Front 1:Back
	D09	0:Upper arm 1: Lower arm
	D10	0:Flip 1:No flip
	D11	0:R<180deg 1:R>=180deg
	D12	0:T<180deg 1:T>=180deg
	D13	0:S<180de 1:S>=180deg
	D14 - D15	Reserved by manufacturer
	D16 - D21	Tool number (0 - 23)
	D22 - D27	User coordinate number
	D28 - D31	Reserved by manufacturer
rData[1]		(Extended attribute)
Array	**Pulse**	**Cartesian**
rData[2]	1st axis (S) pulse value	X-axis coordinate (unit: micron)
rData[3]	2nd axis (L) pulse value	Y-axis coordinate (unit: micron)
rData[4]	3rd axis (U) pulse value	Z-axis coordinate (unit: micron)
rData[5]	4th axis (R) pulse value	Wrist angle Rx (unit: 0.0001deg)
rData[6]	5th axis (B) pulse value	Wrist angle Ry (unit: 0.0001deg)
rData[7]	6th axis (T) pulse value	Wrist angle Rz (unit: 0.0001deg)
rData[8]	7th axis(E) pulse value	angle Re (unit: 0.0001deg)
rData[9]	8th axis pulse value	8th axis pulse value (micron in the case of traveling axis)

■ **Return value**

0 : Normal end

Other than 0 : Error

■ Relevant parameter

Number	Function	Setting value	Default
S2C541	Permission of variable and I/O writing during play mode	0: Permitted 1: Not permitted	1 (Not permitted)
S2C542	Permission of variable and I/O writing during edit lock	0: Permitted 1: Not permitted	1 (Not permitted)

NOTICE

When “0” is set to “S2C541”, writing operation during play back mode becomes valid. However, please operate the manipulator with caution since this operation may influence the cycle time.

NOTICE

The “edit lock status” to which S2C542 can specify are following status.

- In alarm status
- External memory is being used
- Data transmitting function is being used
- Specified input EDIT_LOCK (#40064) is turned ON.

YppCancelError

Releases the error status.

■ **Syntax**
int YppCancelError(out ushort ErrorNo)

■ **Parameter**
[out] *ErrorNo*

Error number

Value	Description
0x0000	Normal end

■ **Return value**
0 : Normal end

Other than 0 : Error

YppResetAlarm

Resets the alarm

■ **Syntax**
int YppResetAlarm(out ushort ErrorNo)

■ **Parameter**
[out] *ErrorNo*

Error number

Value	Description
0x0000	Normal

■ **Return value**
0 : Normal end

Other than 0 : Error

YppSetCycle

Sets the cycle mode.

■ **Syntax**
int YppSetCycle(short Cycle, out ushort ErrorNo)

■ **Parameter**
[in] *Cycle*

Cycle

Value	Description
1	Step
2	One Cycle
3	Auto

[out] *ErrorNo*

Error number

Value	Description
0x0000	Normal
0x2060	In alarm status

■ **Return value**
0 : Normal end

Other than 0 : Error

YppSetServoPower

Sets ON/OFF of the servo power.

■ **Syntax**
int YppSetServoPower(short Power, out ushort ErrorNo)

■ **Parameter**
[in] *Power*

Servo power

Value	Description
0	Servo Power OFF
1	Servo Power ON

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2060	In alarm/error status (The case of Servo ON only)
0x3450	Servo power ON failed

■ **Return value**
0 : Normal end

Other than 0 : Error

YppSetMasterJob

Registers a specified job as a master job.

Syntax

int YppSetMasterJob(short TaskNo, string JobName, out ushort ErrorNo)

Parameter

[in] *TaskNo*

Task number

Value	Description
0	Master Task
1	SubTask 1
2	SubTask 2
3	SubTask 3
4	SubTask 4
5	SubTask 5

[in] *JobName*

Job name (up to 32 characters for a job name)

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2060	In alarm/error status
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

Return value

0 : Normal end

Other than 0 : Error

YppSetCurrentJob

Set the specified job and line number as the current job.

Syntax

int YppSetCurrentJob(ushort LineNo, string JobName, out ushort ErrorNo)

Parameter

[in] *LineNo*

Line number of the job

[in] *JobName*

Job name (up to 32 characters available)

[out] *ErrorNo*

Error number

Member: <*err_no*> error number

Value	Description
0x0000	Normal
0x2010	Manipulator is in operation
0x2060	In alarm/error status
0x2110	Inaccessible job
0x4040	Specified job not found
0x5200	Specified Line No. is out of range

Return value

0 : Normal end

Other than 0 : Error

YppStartJob

Registers a specified job as a master job and starts it.

Syntax

int YppStartJob(short TaskNo, string JobName, out ushort ErrorNo)

Parameter

[in] *TaskNo*

Task number

Value	Description
0	Master Task
1	SubTask 1
2	SubTask 2
3	SubTask 3
4	SubTask 4
5	SubTask 5

[in] *JobName*

Job name (up to 32 characters for a job name)

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0x2080	In TEACH mode status
0x3040	The origin position is not registered
0x3050	Out of range (ABSO data)
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

Return value

0 : Normal end

Other than 0 : Error

YppHold

Turns ON/OFF the hold function.

■ **Syntax**
int YppHold(short Hold, out ushort ErrorNo)

■ **Parameter**
[in] *Hold*

Hold status

Value	Description
0	Hold OFF
1	Hold ON

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded

■ **Return value**
0 : Normal end

Other than 0 : Error

YppWaitForJobEnd

Waits for the job completion or expiration of specified time duration.

Syntax

int YppWaitForJobEnd(short TaskNo, short WiatTime, out ushort ErrorNo)

Parameter

[in] *TaskNo*

Task number

Value	Description
0	Master Task
1	SubTask 1
2	SubTask 2
3	SubTask 3
4	SubTask 4
5	SubTask 5

[in] *WaitTime*

Waiting time

Note: If “-1” is specified, waits unlimitedly

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0xFFFF	Other error

Return value

0 : Normal end

Other than 0 : Error

YppDeleteJob

Deletes the specified job.

Syntax

int YppDeleteJob(string JobName, out ushort ErrorNo)

Parameter

[in] *JobName*

Job name (up to 32 characters available)

[out] *ErrorNo*

Error number

Value	Description
0x0000	Normal
0x2010	Manipulator is in operation or cannot delete JOBs during the robot operation
0x2060	In alarm/error status
0x2080	Play mode
0x2090	Accessing to the specified job
0x4020	Edit lock job
0x4040	Specified JOB not found

Return value

0 : Normal end

Other than 0 : Error

YppSetSVarInfo

Sets the value of S variable

■ **Syntax**
int YppSetVarData(ushort DataIndex, string SVarData)

■ **Parameter**
[in] *DataIndex*

Variable number

[in] *SVarData*

S variable value (S variable: 32 characters at maximum) (set value)

■ **Return value**

0	: Normal end
Other than 0	: Error

■ Relevant parameter

Number	Function	Setting value	Default
S2C541	Permission of variable and I/O writing during play mode	0: Permitted 1: Not permitted	1 (Not permitted)
S2C542	Permission of variable and I/O writing during edit lock	0: Permitted 1: Not permitted	1 (Not permitted)

NOTICE

When “0” is set to “S2C541”, writing operation during play back mode becomes valid. However, please operate the manipulator with caution since this operation may influence the cycle time.

NOTICE

The “edit lock status” to which S2C542 can specify are following status.

- In alarm status
- External memory is being used
- Data transmitting function is being used
- Specified input EDIT_LOCK (#40064) is turned ON.

YppSetJogCtrlGroup

Switches the control group of the jog operation target.

■ **Syntax:**
int YppSetJogCtrlGroup(uint GroupKind, uint GroupNo, out ushort ErrorNo);

■ **Parameter**
[*in*] GroupKind

Control group kind

Value	Description
1	ROBOT
2	BASE
3	STATION

[*in*] GroupNo

Control group number

[*out*] ErrorNo

Error number

Value	Description
0x2010	Control group is in operation.
0x2060	In alarm / error status
0x2080	Not in TEACH mode status
0x3370	Specified control group error

■ **Return value**
0 : Normal end

Other than 0 : Error

YppSetToolNo

Sets the operation tool number to the specified control group.

■ **Syntax:**
int YppSetToolNo(uint RobotNo, short ToolNo, out ushort ErrorNo);

■ **Parameter**
[*in*] RobotNo

Robot number for acquiring the tool number

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)

[*in*] ToolNo

Operation tool number

[*out*] ErrorNo

Error number

Value	Description
0x2010	Control group is in operation.
0x2060	In alarm / error status
0x2080	Not in TEACH mode status
0x3370	Specified control group error
0x3470	Specified tool number error (during the jog operation, the tool switching function is invalid, while the tool selection window appears)

■ **Return value**
0 : Normal end

Other than 0 : Error

5 MOTION CONTROL API

YppIMOV

Moves the current position of the manipulator with the incremental value of linear motion.

Syntax

int YppIMOV(uint Ctrlgrp, int Speed, short VType, short Frame, short ToolNo, int[] Pos, out ushort ErrorNo)

Parameters

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[in] *Speed*

Motion speed Unit : micron/s, 0.0001deg/s

[in] *VType*

Selection of motion speed

Value	Description
0	Control point
1	Position angular

[in] *Frame*

Coordinate system ID

Value	Description
0	Base coordinate
1	Robot coordinate
2, 3, …	User coordinate 1, 2, …

[in] *ToolNo*

Tool number 0 to 63

[in] *Pos (int[12])*

Specified position (Setting Value) ※Requires area for int[12]

Robot axis

Value	Description
Pos[0]	X-axis coordinate (unit: micron)
Pos[1]	Y-axis coordinate (unit: micron)
Pos[2]	Z-axis coordinate (unit: micron)
Pos[3]	Wrist angle Rx (unit: 0.0001deg)
Pos[4]	Wrist angle Ry (unit: 0.0001deg)
Pos[5]	Wrist angle Rz (unit: 0.0001deg)
Pos[6]	angle Re (unit: 0.0001deg)
Pos[7]	(Reserved)
Pos[8]	(Not in use)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

External axis

Value	Description
Pos[0]	(Not in use)
Pos[1]	(Not in use)
Pos[2]	(Not in use)
Pos[3]	(Not in use)
Pos[4]	(Not in use)
Pos[5]	(Not in use)
Pos[6]	1st external axis pulse value (micron in the case of traveling axis)
Pos[7]	2nd external axis pulse value (micron in the case of traveling axis)
Pos[8]	3rd external axis pulse value (micron in the case of traveling axis)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

Set "0" for data Pos[8] to Pos[11] if the system has no external axis.

Set “0” for data Pos[0] to Pos[5] and Pos[9] to Pos[11] if the system has any external axes.

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0x2080	In TEACH mode status
0x3040	The origin position is not registered
0x3050	Out of range (ABSO data)
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

■ **Return value**

0 : Function succeeded.

Nonzero : Function failed.

YppMOVJ

Moves the manipulator to the specified position with the joint motion.

Syntax

int YppMOVJ(uint CtrlGrp, int Speed, short Frame, short Config, short ToolNo, int[] Pos, out ushort ErrorNo)

Parameters

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[in] *Speed*

Motion speed 1 to 10000 (Unit : 0.01%) ※1 - 10000 representing 0.01 to 100.0%

[in] *Frame*

Coordinate system ID

Value	Description
0	Base coordinate
1	Robot coordinate
2, 3, …64	User coordinate 1, 2, …63

[in] *Config*

Configuration

Value	Description	
D00	0:Front	1:Back
D01	0:Upper arm	1:Lower arm
D02	0:Flip	1:No flip
D03	0:R<180deg	1:R>=180deg

Value	Description
D04	0:T<180deg 1:T>=180deg
D05	0:S<180deg 1:S>=180deg
D06-D15	Reserved

[in] *ToolNo*

Tool number 0 to 63

[in] *Pos (int[12])*

Specified position (Setting Value) ※Requires area for int[12]

Robot axis

Value	Description
Pos[0]	X-axis coordinate (unit: micron)
Pos[1]	Y-axis coordinate (unit: micron)
Pos[2]	Z-axis coordinate (unit: micron)
Pos[3]	Wrist angle Rx (unit: 0.0001deg)
Pos[4]	Wrist angle Ry (unit: 0.0001deg)
Pos[5]	Wrist angle Rz (unit: 0.0001deg)
Pos[6]	angle Re (unit: 0.0001deg)
Pos[7]	(Reserved)
Pos[8]	(Not in use)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

External axis

Value	Description
Pos[0]	(Not in use)
Pos[1]	(Not in use)
Pos[2]	(Not in use)
Pos[3]	(Not in use)
Pos[4]	(Not in use)
Pos[5]	(Not in use)
Pos[6]	1st external axis pulse value (micron in the case of traveling axis)
Pos[7]	2nd external axis pulse value (micron in the case of traveling axis)
Pos[8]	3rd external axis pulse value (micron in the case of traveling axis)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

Set "0" for data Pos[8] to Pos[11] if the system has no external axis.

Set “0” for data Pos[0] to Pos[5] and Pos[9] to Pos[11] if the system has any external axes.

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0x2080	In TEACH mode status
0x3040	The origin position is not registered
0x3050	Out of range (ABSO data)
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

■ **Return value**

0 : Function succeeded.

Nonzero : Function failed.

YppMOVL

Moves the manipulator to the specified position with the linear motion.

Syntax

int YppMOVL(uint CtrlGrp, int Speed, short VType, short Frame, short Config, short ToolNo, int[] Pos, out ushort ErrorNo)

Parameters

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[in] *Speed*

Motion speed Unit : micron/s, 0.0001deg/s

[in] *VType*

Selection of motion speed

Value	Description
0	Control point
1	Position angular

[in] *Frame*

Coordinate system ID

Value	Description
0	Base coordinate
1	Robot coordinate
2, 3, …	User coordinate 1, 2, …

[in] *Config*

Configuration

Value	Description
D00	0:Front 1:Back
D01	0:Upper arm 1:Lower arm
D02	0:Flip 1:No flip
D03	0:R<180deg 1:R>=180deg
D04	0:T<180deg 1:T>=180deg
D05	0:S<180deg 1:S>=180deg
D06-D15	Reserved

[in] *ToolNo*

Tool number 0 to 63

[in] *Pos (int[12])*

Specified position (Setting Value) ※Requires area for int[12]

Robot axis

Value	Description
Pos[0]	X-axis coordinate (unit: micron)
Pos[1]	Y-axis coordinate (unit: micron)
Pos[2]	Z-axis coordinate (unit: micron)
Pos[3]	Wrist angle Rx (unit: 0.0001deg)
Pos[4]	Wrist angle Ry (unit: 0.0001deg)
Pos[5]	Wrist angle Rz (unit: 0.0001deg)
Pos[6]	angle Re (unit: 0.0001deg)
Pos[7]	(Reserved)
Pos[8]	(Not in use)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

External axis

Value	Description
Pos[0]	(Not in use)
Pos[1]	(Not in use)
Pos[2]	(Not in use)
Pos[3]	(Not in use)
Pos[4]	(Not in use)
Pos[5]	(Not in use)
Pos[6]	1st external axis pulse value (micron in the case of traveling axis)
Pos[7]	2nd external axis pulse value (micron in the case of traveling axis)

Value	Description
Pos[8]	3rd external axis pulse value (micron in the case of traveling axis)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

Set "0" for data Pos[8] to Pos[11] if the system has no external axis.

Set “0” for data Pos[0] to Pos[5] and Pos[9] to Pos[11] if the system has any external axes.

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0x2080	In TEACH mode status
0x3040	The origin position is not registered
0x3050	Out of range (ABSO data)
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

■ **Return value**

0 : Function succeeded.

Nonzero : Function failed.

YppPulseMOVJ

Moves the manipulator to the specified pulse position with the joint motion.

Syntax

int YppPulseMOVJ(uint CtrlGrp, int Speed, short ToolNo, int[] Pos, out ushort ErrorNo)

Parameters

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[in] *Speed*

Motion speed 1 to 10000 (Unit : 0.01%) ※1 - 10000 representing 0.01 to 100.0%

[in] *ToolNo*

Tool number 0 to 63

[in] *Pos (int[12])*

Specified position (Setting Value) ※Requires area for int[12]

Robot axis

Value	Description
Pos[0]	1st axis (S) pulse value
Pos[1]	2nd axis (L) pulse value
Pos[2]	3rd axis (U) pulse value
Pos[3]	4th axis (R) pulse value
Pos[4]	5th axis (B) pulse value
Pos[5]	6th axis (T) pulse value
Pos[6]	7th axis (E) pulse value
Pos[7]	8th axis pulse value
Pos[8]	(Not in use)

Value	Description
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

External axis

Value	Description
Pos[0]	(Not in use)
Pos[1]	(Not in use)
Pos[2]	(Not in use)
Pos[3]	(Not in use)
Pos[4]	(Not in use)
Pos[5]	(Not in use)
Pos[6]	(Not in use)
Pos[7]	(Not in use)
Pos[8]	1st external axis pulse value
Pos[9]	2nd external axis pulse value
Pos[10]	3rd external axis pulse value
Pos[11]	4th external axis pulse value

Set "0" for data IPos[8] to IPos[11] if the system has no external axis.

Set “0” for data Pos[0] to Pos[5] if the system has any external axes.

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0x2080	In TEACH mode status
0x3040	The origin position is not registered
0x3050	Out of range (ABSO data)
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

■ **Return value**

0	: Function succeeded.
Nonzero	: Function failed.

YppPulseMOVL

Moves the manipulator to the specified pulse position with the linear motion.

Syntax

YppPulseMOVL(uint CtrlGrp, int Speed, short VType, short ToolNo, int[] Pos, out ushort ErrorNo)

Parameters

[in] *CtrlGrp*

Control group

Value	Description
0	R1 (Robot 1)
1	R2 (Robot 2)
8	B1 (Base 1)
9	B2 (Base 2)
16	S1 (Station 1)
17	S2 (Station 2)
18	S3 (Station 3)

[in] *Speed*

Motion speed 1Unit : micron/s, 0.0001deg/s

[in] *VType*

Selection of motion speed

Value	Description
0	Control point
1	Position angular

[in] *ToolNo*

Tool number 0 to 63

[in] *Pos (int[12])*

Specified position (Setting Value) ※Requires area for int[12]

Robot axis

Value	Description
Pos[0]	1st axis (S) pulse value
Pos[1]	2nd axis (L) pulse value
Pos[2]	3rd axis (U) pulse value
Pos[3]	4th axis (R) pulse value
Pos[4]	5th axis (B) pulse value
Pos[5]	6th axis (T) pulse value
Pos[6]	7th axis (E) pulse value
Pos[7]	8th axis pulse value
Pos[8]	(Not in use)
Pos[9]	(Not in use)
Pos[10]	(Not in use)
Pos[11]	(Not in use)

External axis

Value	Description
Pos[0]	(Not in use)
Pos[1]	(Not in use)
Pos[2]	(Not in use)
Pos[3]	(Not in use)
Pos[4]	(Not in use)
Pos[5]	(Not in use)
Pos[6]	(Not in use)
Pos[7]	(Not in use)
Pos[8]	1st extermal axis pulse value
Pos[9]	2nd external axis pulse value
Pos[10]	3rd external axis pulse value
Pos[11]	4th external axis pulse value

Set "0" for data lPos[8] to lPos[11] if the system has no external axis.

Set “0” for data lPos[0] to lPos[5] if the system has any external axes.

[out] *ErrorNo*

Error number

Value	Description
0x0000	Succeeded
0x2010	Robot is in operation
0x2030	In HOLD status (Pendant)
0x2040	In HOLD status (External)
0x2050	In HOLD status (Command)

Value	Description
0x2060	In alarm/error status
0x2070	In SERVO OFF status
0x2080	In TEACH mode status
0x3040	The origin position is not registered
0x3050	Out of range (ABSO data)
0x3400	Cannot operate MASTER JOB
0x3410	In TEACH LOCK status; or the JOB name is already registered
0x4040	Specified JOB not found

■ **Return value**

0 : Function succeeded.

Nonzero : Function failed.

6 FILE TRANSFER API

YppLoadFile

Loads the specified file.

■ Syntax

int YppLoadFile(int Media, string LoadPath, string FileName)

■ Parameter

[in] *Media*

Medias in which loading files exist.

Value	Description
1	SD Card (PP)
2	USB memory (PP)

[in] *loadPath*

Pointer to the folder name in which loading files exist.

- When the file exists in the route folder of the specified media, specify an empty character string (“”).
- When the number of hierarchy of the holder is two or more, use “\\” to partition them.
  (Example: aaa\\bbb)

[in] *fileName*

Name of the loading file.

■ Return value

0 : Normal end

Other than 0 : Error

NOTICE

In consideration of the program migration from the NX100, by specifying 0 (IMedia=0) to the first parameter, specification with the full path name, which is same as the full path name of NX100, to savePath is enabled.

However, on the other hand, it is recommended to specify “1” or ”2” (IMedia=1 or 2) and use the SD Card (PP) or the USB memory for loading/saving the file.

YppSaveFile

Saves the specified file.

Syntax

int YppSaveFile(int Media, string SavePath, string FileName)

Parameter

[in] *IMedia*

Medias for saving files

Value	Description
1	SD Card (PP)
2	USB memory (PP)

[in] *savePath*

Folder name in which the file is saved.

- When saving the file in the route folder of the specified media, specify an empty character string (“”).
- When the hierarchy of the holder is two or more, use “\\” to partition them.
  (Example: aaa\\bbb)

[in] *fileName*

File name to be saved.

Return value

0 : Normal end

Other than 0 : Error

NOTICE

In consideration of the program migration from the NX100, by specifying 0 (IMedia=0) to the first parameter, specification with the full path name, which is same as the full path name of NX100, to savePath is enabled.

However, on the other hand, it is recommended to specify “1” or ”2” (IMedia=1 or 2) and use the SD Card (PP) or the USB memory for loading/saving the file.

YppRefreshFileList

Saves the specified file.

■ **Syntax**
int YppRefreshFileList(short Extension)

■ **Parameter**
[in] *extension*

File type

Value	Description
1	Independent job file (JBI)
2	Related job file (JBR)

■ **Return value**
0 : Normal end

Other than 0 : Error

YppGetFileCount

Acquires the file number in the file list.

- **Syntax**
  int YppGetFileCount()

- **Parameter**
  No parameter

- **Return value**

0	: Normal end
Other than 0	: Error

YppGetFileName

Acquires the file name in the file list.

■ **Syntax**
int YppGetFileName(int Index, out string JobName)

■ **Paramete**
[in] *index*

Index of the acquiring file.

[out] *SavePath*

File name of the specified index (result of acquisition)

■ **Return value**

0	: Normal end
Other than 0	: Error

YRC1000micro OPTIONS
INSTRUCTIONS

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